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Volatile Organic Compounds Monitored in Supply Chain Seafood/Dairy Industries

Abstract:

It is well known that fruit, meat, seafood, and dairy products will spoil over time from a number of various factors such as the imbalance of light, heat, temperature, or oxygen. In this research paper, we focused on the production of volatile organic compounds in both dairy and seafood products during the shipping and delivery of said sectors. The evolution of VOCs and microbial growth in seafood and dairy foods were investigated through multiple resources. Levels of Toluene, Styrene, Acetone, and Ethylbenzene were prevalent across multiple outside studies in seafood, and aldehydes plus other acids in dairy. Ultimately, monitoring VOCs before and during transport will ensure the safety of the food and will allow operators to take action in case of complications in refrigeration equipment.

Introduction:

Volatile Organic Compounds are a group of organic chemicals categorized as carbon-containing chemicals emitted into the air from a liquid or solid state. Some of these "VOC's", as they have come to be known, are extremely harmful to human health and safety. Most can have detrimental effects on both humans and the environment–many are even heavily present in our everyday lives. These compounds are known to be released through certain liquids or solids, like petroleum fuels, cigarettes, automobile exhaust, or plastic household products, but some VOCs have been constantly emitted from organisms for eons. Plants release volatile compounds like terpenoids or green leaf volatiles, as a form of defense, while others are released as a by-product of energy production. Although, plants are not the only organic matter that releases VOCs. Both while living and dead, all organisms continuously produce waste that breakdown into the processes of volatile compound emission. As soon as an animal is killed, it immediately begins the process of spoilage, leading to another amass of VOC production. Today, there is major concern surrounding the discussion of volatile compounds and microbial activity in international food products. One area of interest is the seafood and dairy sector because of its high demand in today's economy. Upon preparation for shipment, seafood can be preserved in numerous ways, including but not limited to: canning, drying, smoking, salting, or even pickling. But the most common method of preservation is freezing, as colder temperatures slow down the matter decomposition and thus inhibit the bacterial growth and VOC emission in food. Like seafood, dairy products are best shipped frozen or cold because the low-temperature curb spoilage effects within. To help maintain this principle in long-term supply chain distribution, there are two main methods of transportation: aircraft and ship. Reefer ships transport all refrigerated goods and perishables in refrigeration units across the water. Units that include refrigeration, temperature monitoring, compressors, and insulation all to ensure the content's protection. Similar to reefer ships, airline ships, or perishable air food freight flights, are equipped with refrigeration units known as "ULDs(unit load devices)". Units are lined with refrigeration or filled with dry ice to keep a temperature of anywhere from 20 to -20 degrees Celsius. Although many regulations are in place surrounding sellers, ultimately, low temperatures only slow down the VOC production in products. Air freights are faster and more reliable but tend to be extremely pricey, meaning most perishables are transported through water. Couple with distant shipping nations, perishable shipping spoilage has become a major area of concern surrounding the food industries

1. Methods

1.1 Colorimetric dye-based sensors

Markers and indicators that change color, based on external stimuli(s) have been investigated and in application for years. Mainly these Colorimetric dye-based sensors are utilized similarly to other gas sensors but are cheap, inexpensive, and efficient. These sensors are able to change color in response to pH changes and have effectively been in use to monitor quality changes in our food. (Fang, 2022) effectively demonstrated the ability to determine spoilage levels in shrimp using colorimetric sensors. White-legged Shrimp samples were homogenized, placed beneath a solution-soaked sensor paper, and averaged for a "color-reading". Shrimp samples were then placed under GC-M/S to identify volatile compounds and thus the two readings from both techniques were matched. Similarly, (Sarnoski, 2008) homogenized crab samples and analyzed those with the help of colorimetric sensors to calculate the ammonia concentration. These readings along with Gas chromatography measurements were vital to determine the freshness and quality of blue crab. However, one must consider every variable, such as time, temperature, or environment to fully understand the results from colorimetric dye-based sensors

1.2 Gas & Liquid Chromatography

Measuring VOCs from the headspace of food products is a very well-established and accurate method to determine substances within samples. GC-M/S (GC Mass Spectrometry) and GC liquid chromatography are expensive laboratory equipment that require trained technicians to operate. However, various forms of GC are perfect and accurate instruments to detect various VOCs in a given spoilage sample. GCs are non-cost effective instruments commonly used in detecting spoilage for the purpose of continuous monitoring during transportation and storage(Putri, 2022). These machines are perfect for understanding the makeup of our everyday substances.

2. Results

2.1 Seafood

Seafood spoilage, whether that be in fish, shrimp, crab, or any similar type of seafood, almost always results from three primary mechanisms: Microbial growth, Enzymatic autolysis, and lipid oxidation. As soon as an organism is slaughtered, it begins to spoil. After death, microbial bacteria begin to invade tissue and muscle onboard the fish. Specific Spoilage organisms, or "SSO's", are specific to the environment of the pre-mortem organisms. Meaning SSO's from a warmer environment will differ from that of a colder one. Pseudomonas and Aeromonas are regular for chilled fish whilst Shewanella spp. is a typical SSO for fish from temperate water. Enzymatic autolysis is caused by the self-breakdown of fish tissue. Upon death, all oxygen/energy production comes to a halt and the process of degradation begins to affect the textural quality of the organism. Lastly, because fish is typically a highly saturated fat food, fatty acids are heavily present. This leads to lipid oxidation, a process of molecular displacement in which fatty acids are oxidatively degraded and eventually lead to rancidity in fish. All three of these methods occur often during the spoilage of fish but not all of them are in our area of interest. During harvesting, if fish are stored at 0 degrees Celsius, it can decrease microbial and Enzymatic spoilage but can not prevent oxidation. Therefore our efforts should mostly be within the realm of VOCs as a result of oxidation.

Spoilage is completely dependent on a number of biochemical reactions within foods. Nonetheless, lipid oxidation and microbial spoilage are considered to be the biggest factors that lead to volatile compound production within the seafood sector. In (Kim, 2014) study discussing the composition of odorants from different food types in shipping packaging, samples of decaying Kimchi, fresh, and salted fish were examined on the relationships between the food types and VOC compositions during spoilage. In the end, it was revealed that a majority of aldehydes were produced followed by alcohols, ketones, esters, and finally one acid. In addition, a number of volatile organic compounds (VOCs) including methyl ethyl ketone, toluene, styrene, formaldehyde, and benzene were released at respectively high numbers during food decaying processes.

(Ying-Jie, 2020) study about the effect of Cold Chain Logistics on Lipid Oxidation and VOCs produced similar results as well. Alcoholic volatile compounds like 1-Penten-3-ol and 2-octyn-ol being the amplest in chemical content related to lipid oxidation and aldehydes including Hexanal and Nonanal increasing at high ranges. Heavy microbial activity was found to also reveal volatile compound activity as well. Several VOCs released from common bacteria like Pseudomonas or Pschyobracter were shown to establish off-odor and off-flavor descriptions in studies of undesirable formations in fish during about a 12-day span(Anagnostopoulos, 2011). The result concluded that bacterias, Pseudomonas and Pschyobracter, dominantly produced Penten-3-ol and ethanol along with other VOCs during a red seabream's spoilage. (*Psychrobacter* species are cold-tolerant and often associated with marine environments, further causing opportunistic infections). The examined timeline was effective because it closely matched the period of time a boat ride from Eastern European and Asian countries takes. Both of these geographical areas exist in the realm where the U.S tends to do the most business with seafood and dairy industries. However, another major complication is temperature fluctuations in on and off-board transportation. In recent years, fluctuations have been found to heavily affect and accelerate VOC production inside seafood during study, so maintaining a constant climate is crucial in removing unnecessary factors from affecting spoilage.

Pollutants name		CAS number	Structural formula	Threshold[a] (ppm)	MDL (ng)	Precision[b] (%)
Full name	Short name					
Acetone	AC	67-64-1	CH_3COCH_3	42	10.5	1.9
Methyl ethyl ketone	MEK	78-93-3	$CH_3COCH_2CH_3$	0.44	0.19	0.8
Toluene	T	108-88-3	$C_6H_5CH_3$	0.33	0.18	0.8
Ethylbenzene	EB	100-41-4	$C_6H_5C_2H_5$	0.17	0.18	1.5
m, *p*-xylene	MPX	108-38-3/106-42-3	$(CH_3)_2C_6H_4$	0.041/0.058[c]	0.09	1.1
Styrene	ST	100-42-5	$C_6H_5CH{=}CH_2$	0.035	0.08	1.1
o-xylene	OX	95-47-6	$(CH_3)_2C_6H_4$	0.38	0.13	1.1

MDL method detection limit

[a] Nagata (2003)

[b] Precision is expressed in terms of relative standard error (RSE) of seven replicate analyses

[c] *m*-xylene = 0.041 ppm; *p*-xylene = 0.058 ppm

Ref 6. (The table above shows VOCs in three types of fish during the decaying process)

2.2 Dairy

In contrast, dairy, with a heavily distinct taste, is distinguishable for its off-flavor and off-odor taste that can raise suspicion about VOC presence. Dairy consists of many compounds, but its main components include lactose, protein, fat, and water. During its quality standard processes, the initial cow's milk is sterilized in a process known as pasteurisation, where milk is heated up and then rapidly cooled to kill bacteria and other microbes. Some countries use raw milk, or unpasteurized milk, as their main dairy source, but this milk can carry many foodborne illnesses such as Salmonella. From there, depending on the final product of the dairy it will be fermented and prepared in specific ways. However, for the most part, pasteurized milk spoilage is due to VOC-producing microbial activity and can be directly linked to giving rise to off-odour/tastes. In cheese and yogurt, these volatile compounds are abundant in creating the well-known distinct characteristics that we have come to love today.

A 2018 study using a GC/MS found a total of 47 VOCs belonging to seven chemical groups identified in the spoiled milk. The following seven VOC compounds were found in the spoiled milk: 3-Methylbutan-1-ol(C5H12O), 2 methylpropan-1-ol(C4H10O), 3-hydroxybutan-2-one(C9H18O3), butan-2,3-dione(C4H6O2), butanoic (C4H8O2)and hexanoic (C6H12O2) acids. (Ziyainia, 2019) another 2018 study centered around milk spoilage, was conducted using methods of colorimetric sensors along with GC-MS. The research positively associated milk spoilage with bacterium in all average shelf life(7-19 degrees celsius) temperatures during only a couple day time span. The volatile organic compound profile of cheese is still being investigated today, as many kinds of cheese have close relationships with fermentation processes. Mozzarella, a typical cheese type sold globally, emits various levels of different VOCs including high levels of 3-Methyl butanal, hexanal, Decanal, and Butanoic acid(Natrella, 2020). By further understanding dairy product spoilage, we can save many unnecessary kinds of milk and dairy product ruin.

3. Discussion

Preliminary Cautions

To ensure the safety and health of consumers on the end of the supply chain, additional preliminary cautions should be added in place. Most things can not be done to outright prevent the processes of nature, but it only becomes possible to attempt in further delaying spoilage. While perhaps “better-off” companies might manage to consistently ship products through the air(saving precious time), the majority of companies arguably cannot afford that cost. The best preventive methods are therefore in developing more advanced methods of packaging, keeping monitors inside of the packaging, or even treating perishables prior to shipping. Industries should consider the possibility of adding more checkpoints along the transport as well. Colorimetric dye-based indicators that change based on responses to pH values could set forth an inexpensive, easy, and quick way to establish whether a product satisfies expected quality levels. Developing purpose-built micro GCs to measure target VOCs could additionally ensure food safety across shipment and storage as well. Simply put, trade businesses might evaluate extra practices to guarantee greater protection of food products.

Conclusion

In this study, volatile organic compounds produced during seafood and dairy product spoilage were identified and evaluated. We found that during and even before spoilage, levels of Benzene, Toluene and other VOCs were present in fish and dairy(in the course of a similar shipment timeline to that used in modern freighting). Increases in volatile compound levels in dairy and seafood products have been directly linked to indicators of spoilage. Further, the appearance of several uncommon VOCs found during shipment and storage lead to the detection of food spoilage. Scientists have identified specific VOCs related to seafood and dairy spoilage. Compounds like that of 1-Penten-3-ol, Styrene, Benzene, Toluene, and formaldehyde were found to be linked to seafood, while similar volatiles such as Acetone, and several acids are connected to dairy spoilage. Additionally, the food industry can and should use inexpensive methods to monitor VOC levels during large-scale industrial shipments to detect potential spoilage. Monitoring refrigeration equipment, and food temperature are the key factors used in supply-chain assurance but are insufficient. Because of the existence of certain bacteria levels within food, spoilage can accelerate, even in freezing temperatures. Several final methods to prevent this the use of colorimetric or GC sensors to measure PH or VOC levels, monitoring contents to ensure quality standards.

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